

BrickDisplay: Large, Protruding 3D Image Presentation From Multiple Non-Aligned Monitors With Perceptual Transparency

Yuri Mikawa
The University of Tokyo
Tokyo, Japan

yuri_mikawa@ipc.i.u-tokyo.ac.jp

Taiki Fukiage
NTT Communication Science Laboratories
Kanagawa, Japan
taiki.fukiage@ntt.com



Figure 1: (a) Comparison of methods to present large 3D images. (Top) Conventional method (tilled display) aligns identical monitors, forcing 3D content behind screens due to bezel occlusions. (Bottom) BrickDisplay uses non-aligned monitors of various sizes and employs perceptual transparency (PT) to present 3D images in front. (b) A demonstration of BrickDisplay, where a large 3D dolphin image appears in front of non-aligned monitors by inducing PT with the appropriate image luminance design. (c) Conceptual illustration of how the dolphin appears to protrude when observed via 3D glasses.

Abstract

We propose BrickDisplay, a system that presents large 3D images to seemingly protrude over multiple, non-aligned heterogeneous monitors using affordable, lightweight 3D glasses. Unlike conventional methods, which require aligning multiple homogeneous monitors and restrict 3D content to appear behind them due to bezel occlusion, BrickDisplay enables 3D images to appear in front of and extend beyond the physical gaps and bezels caused by irregular monitor placement. We achieve this by leveraging perceptual transparency (PT)—a visual illusion introduced by appropriately designing the image luminance that makes the content appear semi-transparent and visually override physical obstructions. This approach allows flexible monitor placement without sacrificing the sense of depth. As a result, any available monitors can be assembled to create large-scale, immersive 3D visual experiences with minimal cost and effort.

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1 Introduction

In virtual and augmented reality (VR/AR), large 3D aerial images are desired without heavy head-mounted displays. However, a single cinema-like screen is often impractical due to space and cost constraints. As an alternative, tiled displays that align multiple monitors have been proposed to provide affordable solutions [Sandin et al. 2005] (Fig. 1 (a) left). Nevertheless, as the monitors' bezels act as an occlusion cue by obstructing the image, the depth of 3D images is limited behind the monitors. Additionally, preparing multiple homogeneous monitors and aligning them perfectly is labor-intensive.

To overcome these limitations, we propose BrickDisplay, which arranges monitors of different types and sizes in an irregular layout with gaps between them (Fig. 1 (a) right and (b)). Despite this arrangement, a large 3D image is perceived in front of these distributed monitors beyond bezels. We achieve this by inducing perceptual transparency (PT), an illusion where specific luminance arrangements elicit the impression of a transparent overlay [Anderson 2003]. Specifically, we first calibrate each monitor's posture and apply binocular parallax rendering using perspective projection to ensure a consistent 3D view across monitors through 3D glasses. We then adjust the image's luminance to induce PT, allowing the visual system to interpret the image as semi-transparently overlaid over the gaps and bezels. Furthermore, this configuration induces modal completion—a perceptual process that generates

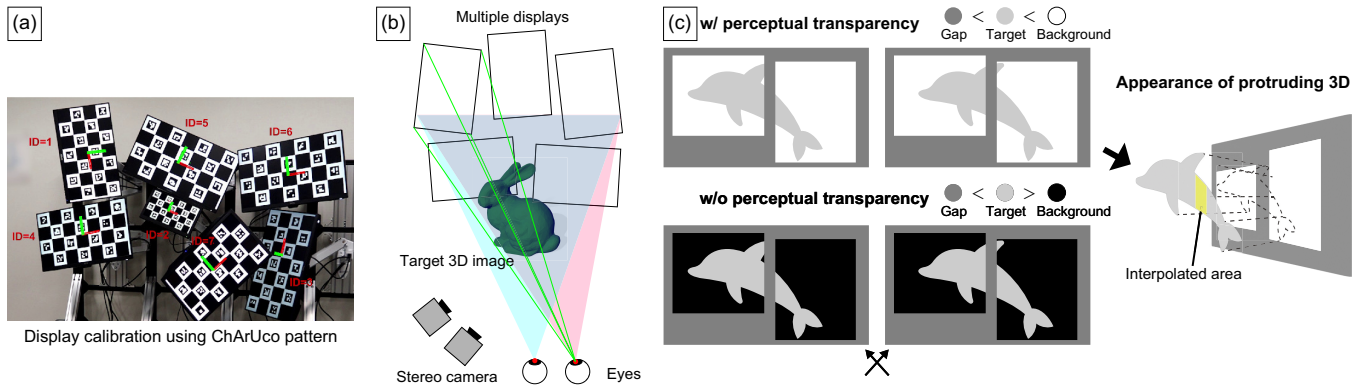


Figure 2: (a) Positional calibration of multiple monitors using ChArUco [Garrido-Jurado et al. 2014]. (b) The positional relationship between each monitor, viewpoint, and stereo camera, which is used to constitute the perspective projection in each monitor. (c) PT can be induced when the target luminance lies between the gap and background luminances [Anderson 2003]. When this condition is met (top), stereoscopic viewing via cross-fusion of the left two images makes the dolphin appear to float in front, with its missing area perceptually filled in (right). Without PT (bottom), the dolphin does not appear in front.

illusory contours to form a continuous shape where physical edges are missing. As a result, viewers perceive a unified 3D object protruding from the distributed monitors, as illustrated in Fig. 1 (c). Although a similar use of PT has also been explored in slit-based 3D displays [Ando et al. 2016], it was confined to a specialized setup. In contrast, BrickDisplay achieves higher-resolution images and broader applicability through flexible monitor arrangements.

BrickDisplay is easily installable and works with any monitor, enabling individuals to create large 3D displays effortlessly. Its ubiquitous design seamlessly integrates smartphones, tablets, monitors, and laptops into a unified screen. Scaling across distributed monitors enables life-sized human and animal presentations, enhancing the intuitive understanding of motion and posture (e.g., sports, dance) while enabling immersive remote interactions with family, artists, and virtual characters.

2 Implementation

System. BrickDisplay consists of multiple monitors of different types and sizes connected to a single or multiple computer(s) equipped with GPUs capable of multiple monitor connections (NVIDIA T1000, 8GB GDDR6 SDRAM). All the monitors are 1920×1080 IPS panels that conserve a wide viewing angle. The size of the monitors ranges from 21.5 to 32 inches. For monitors' positional calibration, a pair of stereo-calibrated Ximea MQ013RG-ON cameras (1280×1024 px) was used to enhance depth accuracy. Gamut calibration across monitors is conducted using a Spyder X Ultra color calibrator.

Positional calibration and 3D rendering across multiple monitors. Positional calibration is essential for consistent 3D presentation across monitors. As shown in Fig. 2 (a), ChArUco [Garrido-Jurado et al. 2014], a pattern enabling the posture estimation and monitor identification, was presented on each monitor and captured by the stereo camera. The estimated posture of each monitor, the rotation and translation, is used to calculate each monitor's view frustum from both eyes' positions, enabling the rendering of binocular parallax images in perspective projection (Fig. 2 (b)). This calibration

ensures a coherent 3D image from the reference viewpoint, while off-axis viewing can reduce continuity when the monitors deviate from a single plane. We use anaglyph rendering as the simplest way to realize stereoscopy on general-purpose monitors with affordable red/cyan 3D glasses. This calibration method by one-shot capture allows adding extra monitors to expand the image presentation area. A limitation is that excessively large inter-monitor gaps reduce the continuity of the perceived 3D image.

Inducing perceptual transparency (PT). In BrickDisplay, PT is utilized to make 3D images protrude beyond the monitors' gap and bezels. To induce PT, the target image luminance should be between the luminances of the gap and background, and the stereoscopy can enhance this effect [Anderson 2003]. The top of Fig. 2 (c) shows a case where this PT condition is met, causing the missing area of the dolphin to appear interpolated and in front. In contrast, the bottom does not meet the PT condition, resulting in no protruding perception. In practice, we apply a linear intensity remapping to the image to ensure that the target image luminance exceeds the gap luminance to properly induce PT. The remapping is objectively conducted based on images acquired using monochrome cameras.

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